

Imagine you have been hired by a small retail business that wants to streamline its operations by creating a new database system. This database will be used to manage inventory, sales, and customer information. The business is a small corner shop that sells a range of groceries and domestic products. It might help to picture your local convenience store and think of what they sell. They also have a loyalty program, which you will need to consider when deciding what tables to create.

Write a 400 to 500-word (in total) essay/report explaining the steps you would take to set up and create this database. Your essay should cover the following points:

1. Understanding the Business Requirements:

- a. What kind of data will the database need to store? Inventory (SKU - product name, product category, price, stock count, **supplier ID**) ; SKU - sales (**transaction ID**, transaction date, quantity, **customer ID** and **supplier ID**); Suppliers SKU (**supplier ID**, supplier name, supplier contact details), customer info SKU (**customer ID**, name, address, contact number, email, membership status, points total), loyalty (**membership no**).*

Note: SKU, in this case are fields in a table.

- b. Who will be the users of the database, and what will they need to accomplish?*

2. Designing the Database Schema:

- a. How would you structure/design the database tables to efficiently store inventory, sales, and customer information? Can you create a simple design it in Word doc.?*
- b. What type of relationship/s between tables are necessary (and can you mention the PK and FK keys) (e.g., how sales relate to inventory and customers)?*

3. Implementing the Database:

- a. What SQL commands would you use to create the database and its tables? Top 4 or 5 queries.*

- b. *Provide examples of SQL statements for creating tables and defining relationships between them. Top 4 or 5.*

4. *Populating the Database:*

- a. *How would you input initial data into the database tables? Can you give examples of SQL INSERT or and VALUES statements.*

5. *Maintaining the Database:*

- a. *What measures would you take to ensure the database remains accurate and up to date?*
- b. *How would you handle backups and data security?*

Your essay/report should include specific examples of SQL commands and explain why each step is necessary for creating a functional and efficient database for the retail business.